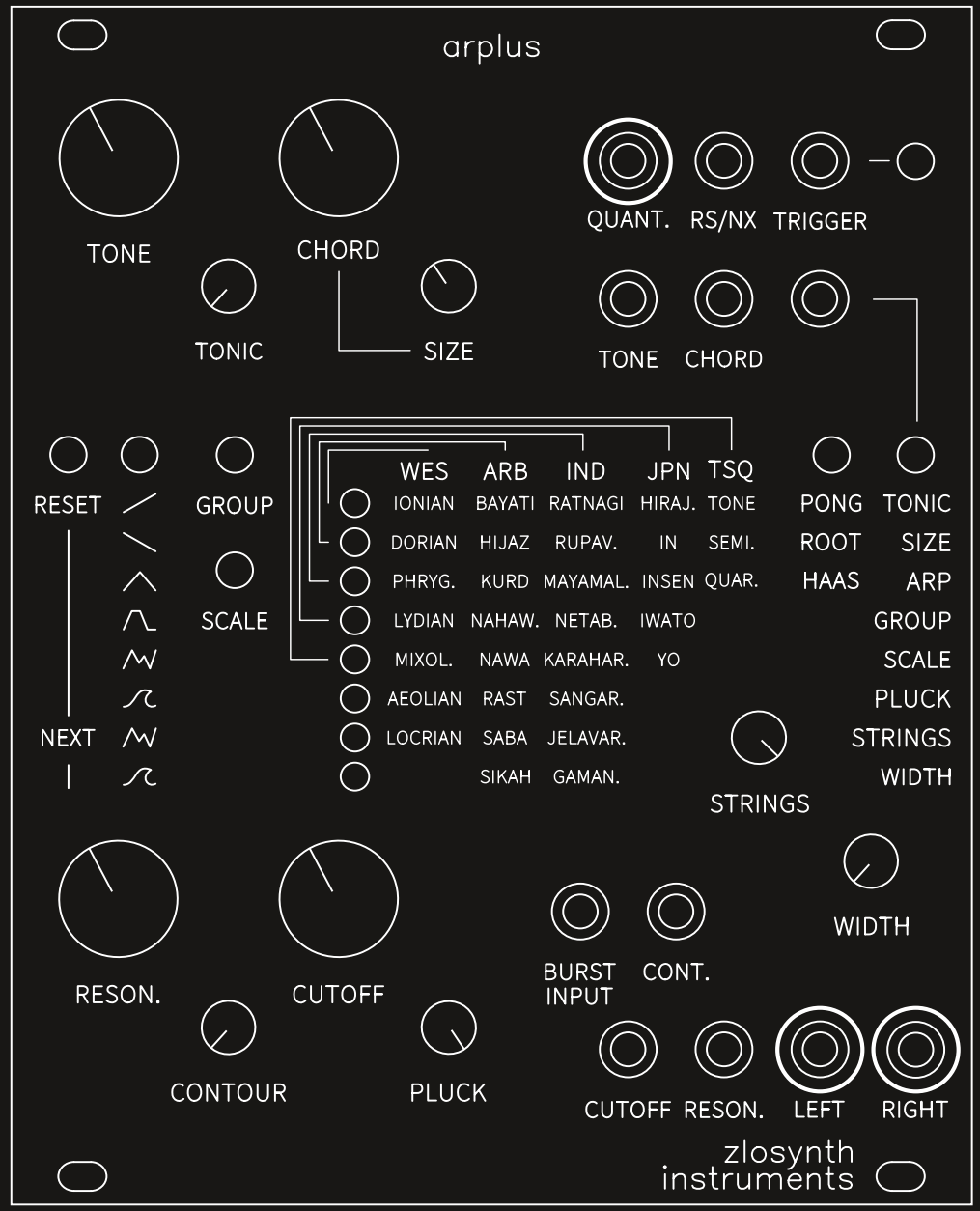
MANUAL

Arplus combines a quantizer, arpeggiator, and string synthesizer into one module. Select from a variety of scales, including Western diatonic scales, microtonal Arabic maqam, Indian melakarta, Japanese pentatonics, or quarter, semi, and whole tone scales. Use the arpeggiator to create note patterns based on a selected chord and scale. Control other modules using 1V/oct output, or take advantage of the built-in 6 channels of string synthesis to produce anything from subtle plucks and staccatos to ensemble-like pads and resonating soundscapes.

Width	20 HP
Depth	28 mm
Power	+12 V (104 mA), −12 V (8 mA)
Input impedance	100 kΩ
Trigger inputs	0 to +5 V, 1 kHz
CV inputs	−5 to +5 V, 16-bit, 1 kHz
CV outputs	0 to +5 V, 16-bit, 1 kHz
Audio	24-bit, 96 kHz

- Karplus-Strong string synthesis with 6-voice polyphony
- External audio input for custom excitation sources
- 31 different scales and chords of up to 8 notes
- Configurable scales
- CV-controllable arpeggiator patterns
- Quantized 1V/oct output
- Three different stereo output modes



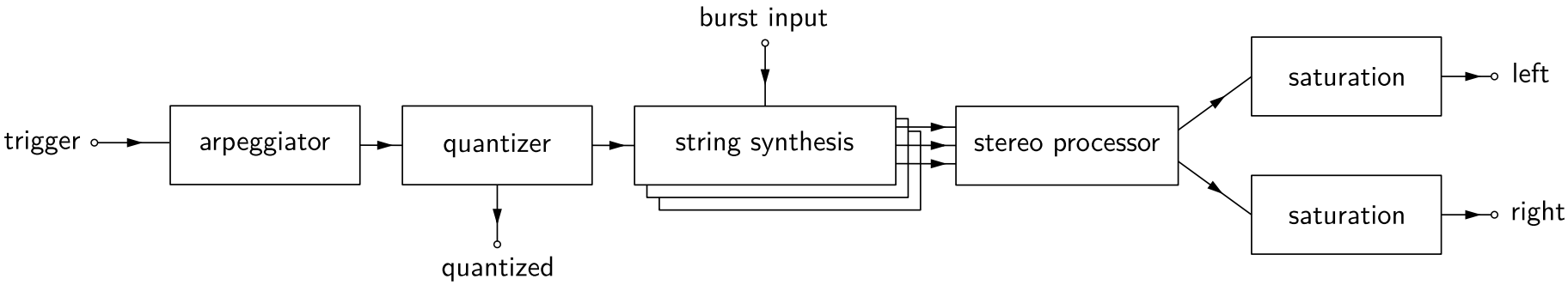
1 Installation

Arplus is 20 HP wide. It is powered by a +12V/−12V 2×5 connector. The red stripe (−12V) has to be connected on the side of the board marked with the white line. The module must be mounted in a eurorack case.

2 Quick start

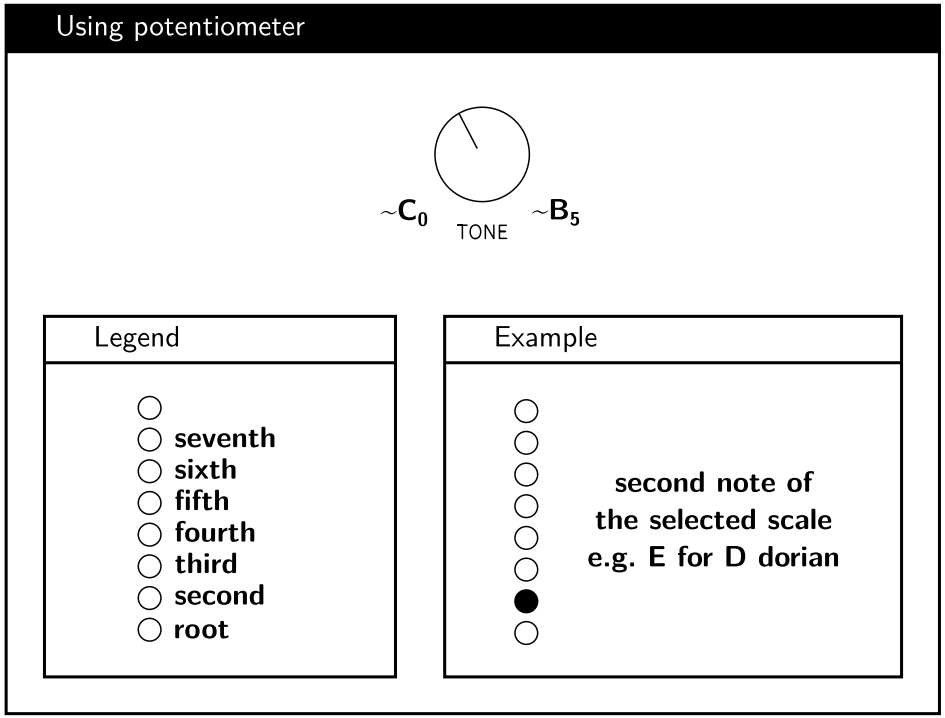
Turn all the knobs to the initial position illustrated on the front page of this manual. Click the ⟋ (ARP type), GROUP, SCALE, and HAAS buttons until each of them lights up the top-most LED. Connect a clock source to the TRIGGER input, and connect the LEFT or RIGHT output to your mixer.

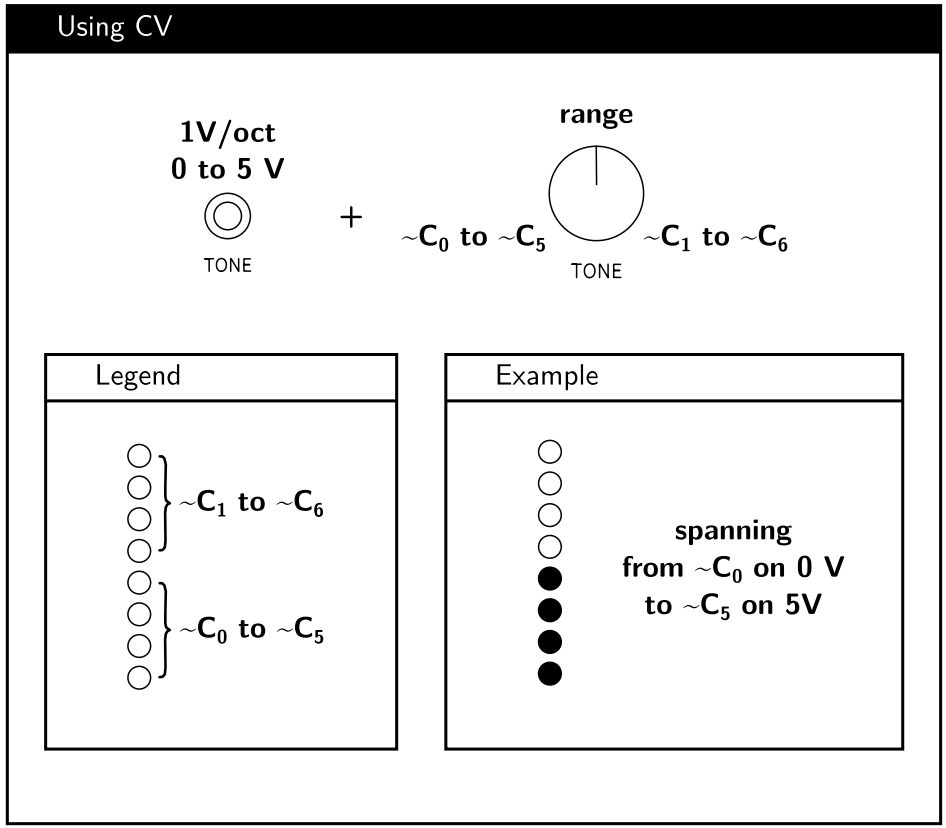
3 Signal flow



4 Root note

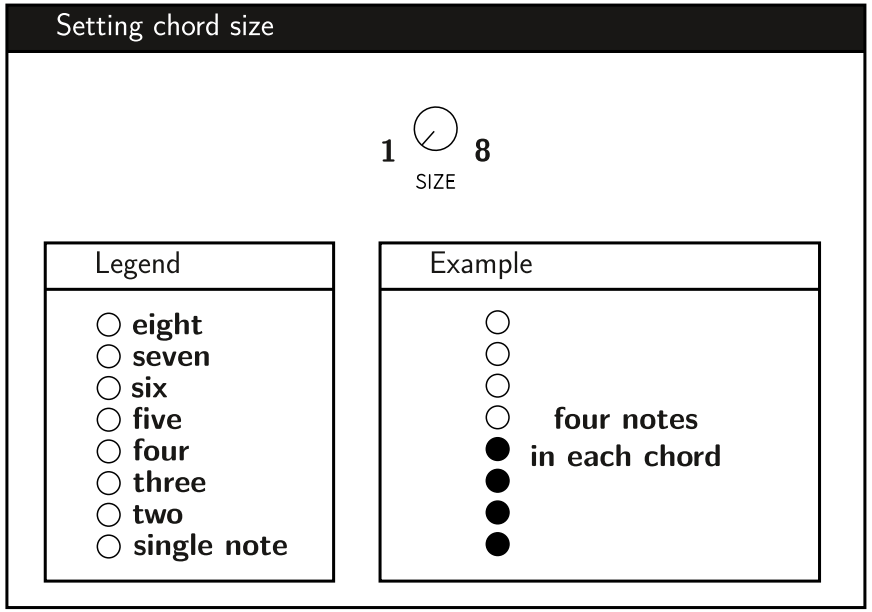
Select the root note of the playing chord. This sets the pitch of the lowest playing tone in the arpeggio.

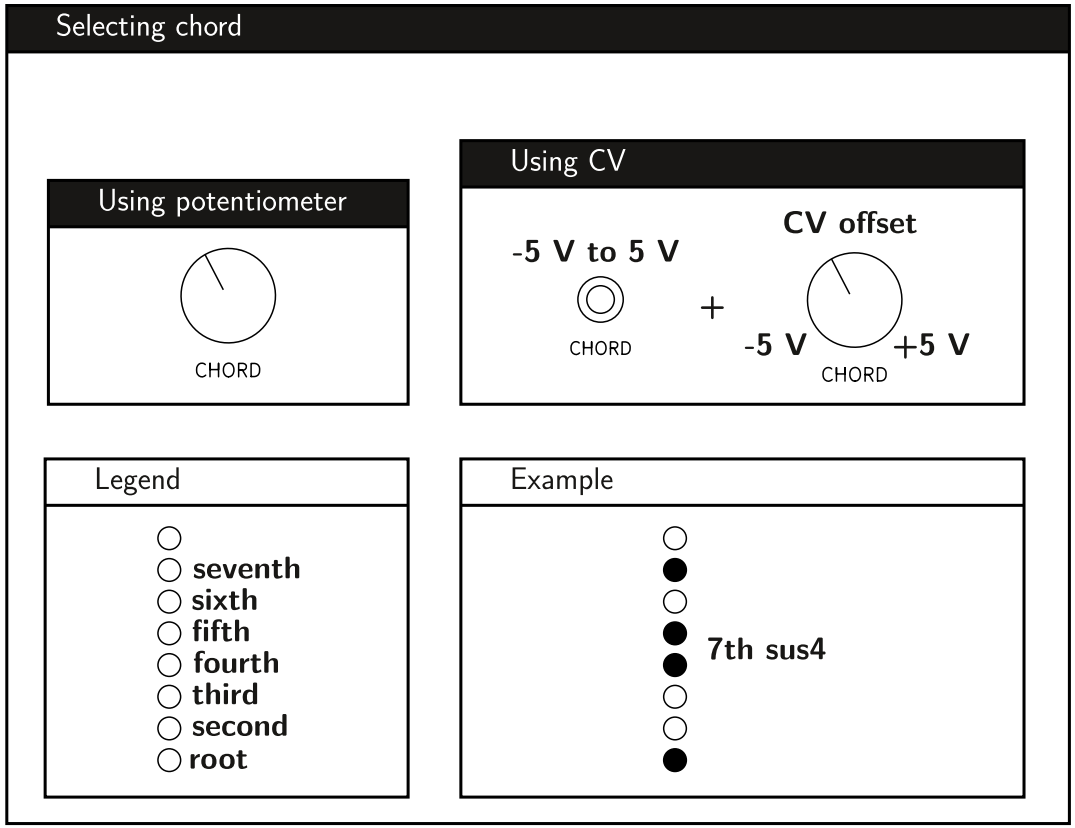




5 Chord

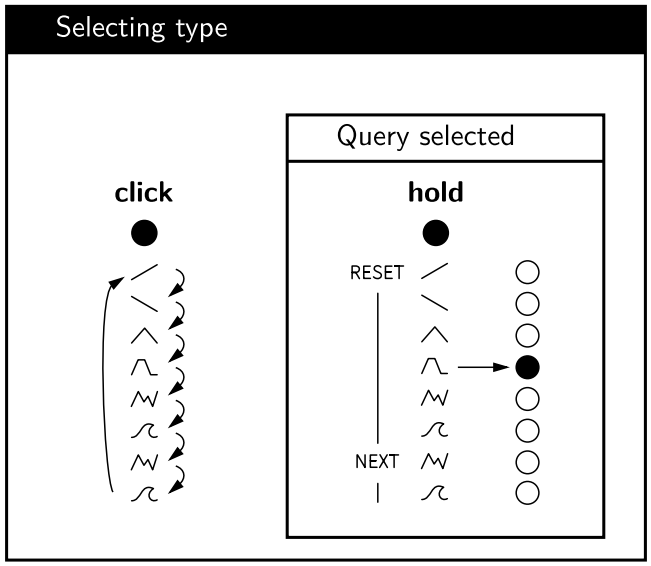
Select from an assorted list chords grouped by size. All the available chords are listed at the end of this manual.

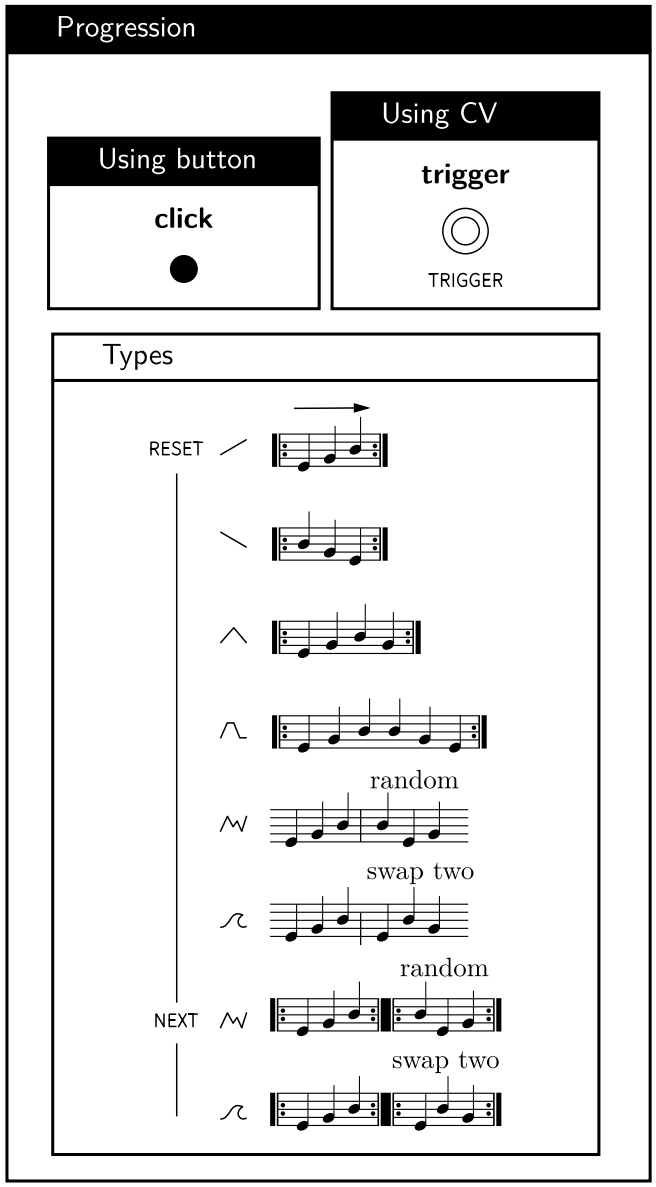


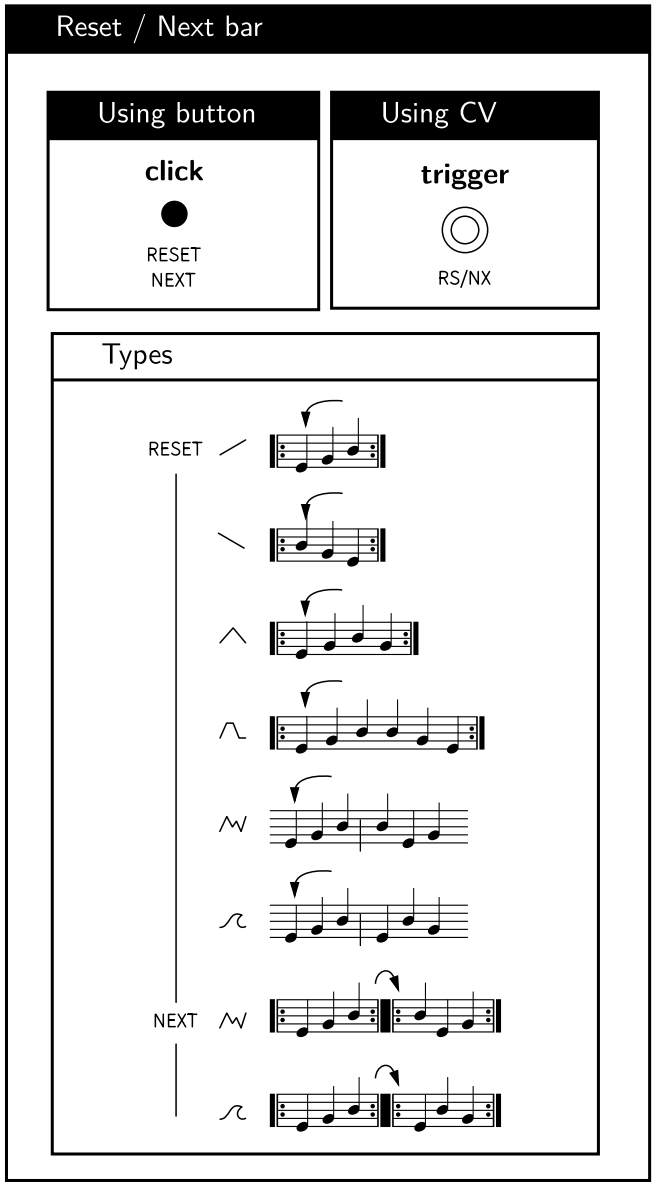


6 Arpeggio

With arpeggio, notes of the selected chord are triggered individually. The selected type defines how should be the notes ordered. Some types can be RESET to their initial position – use two different clock sources for TRIGGER and RS/NX to create interesting patterns. Other types are repeating the same sequence until NEXT is triggered.

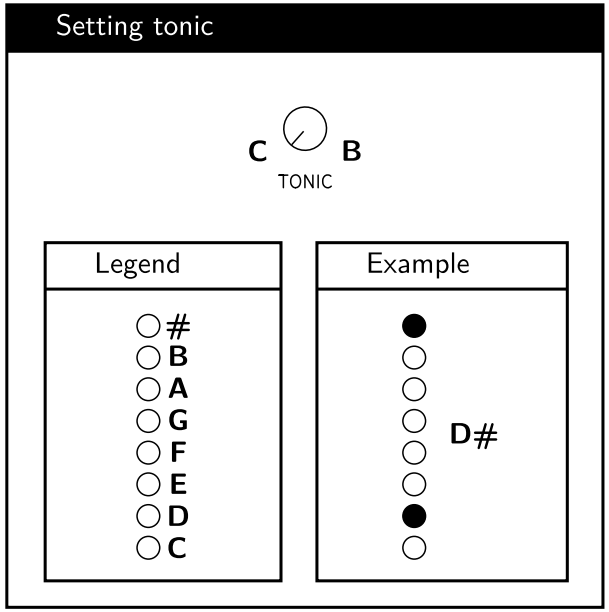


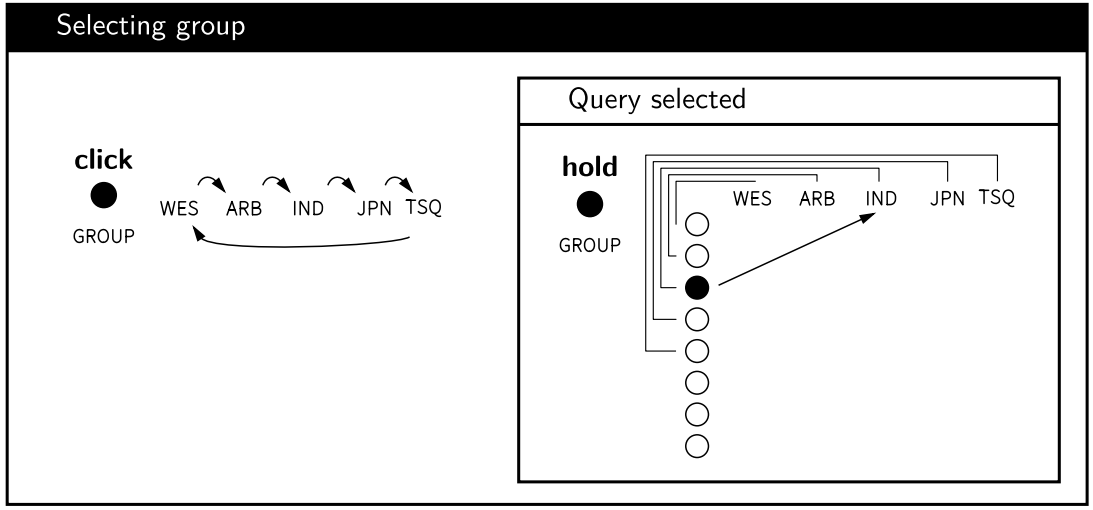


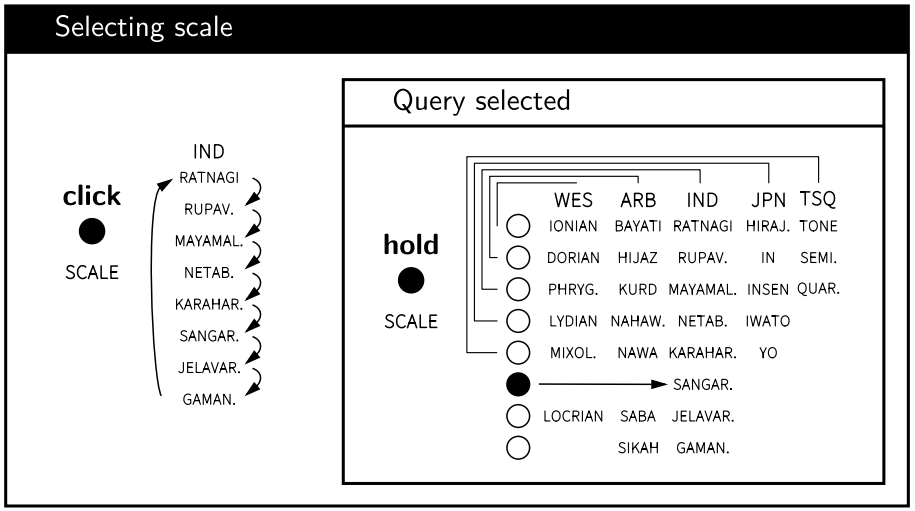


7 Scale

Select one of the 31 available scales. All the available scales are listed at the end of this manual.

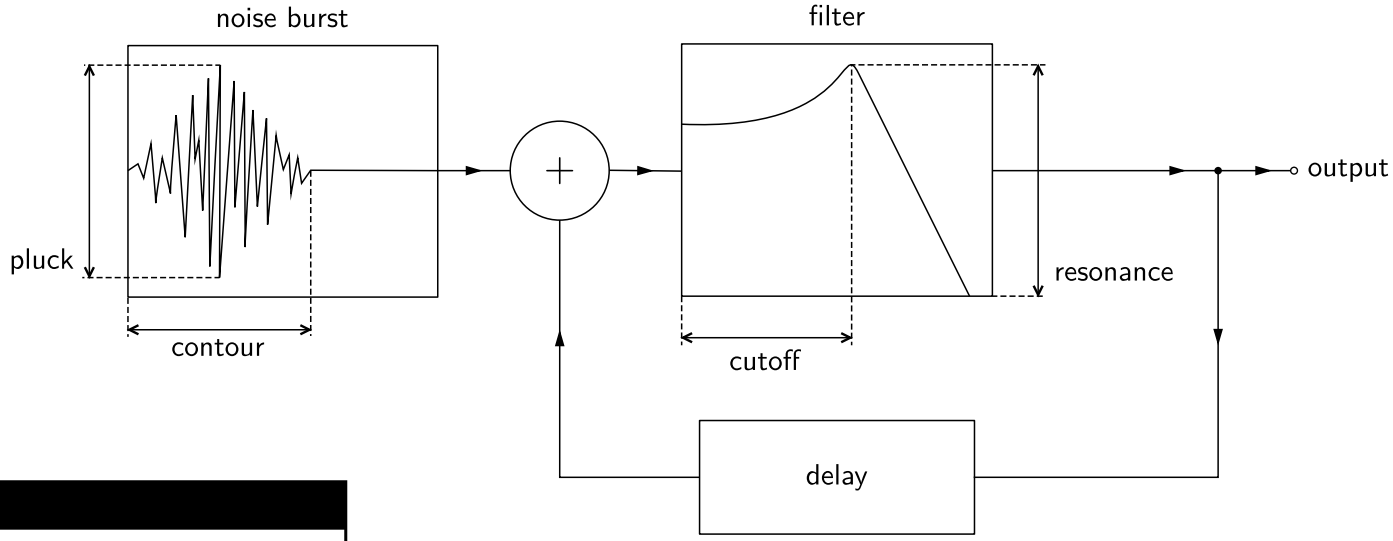


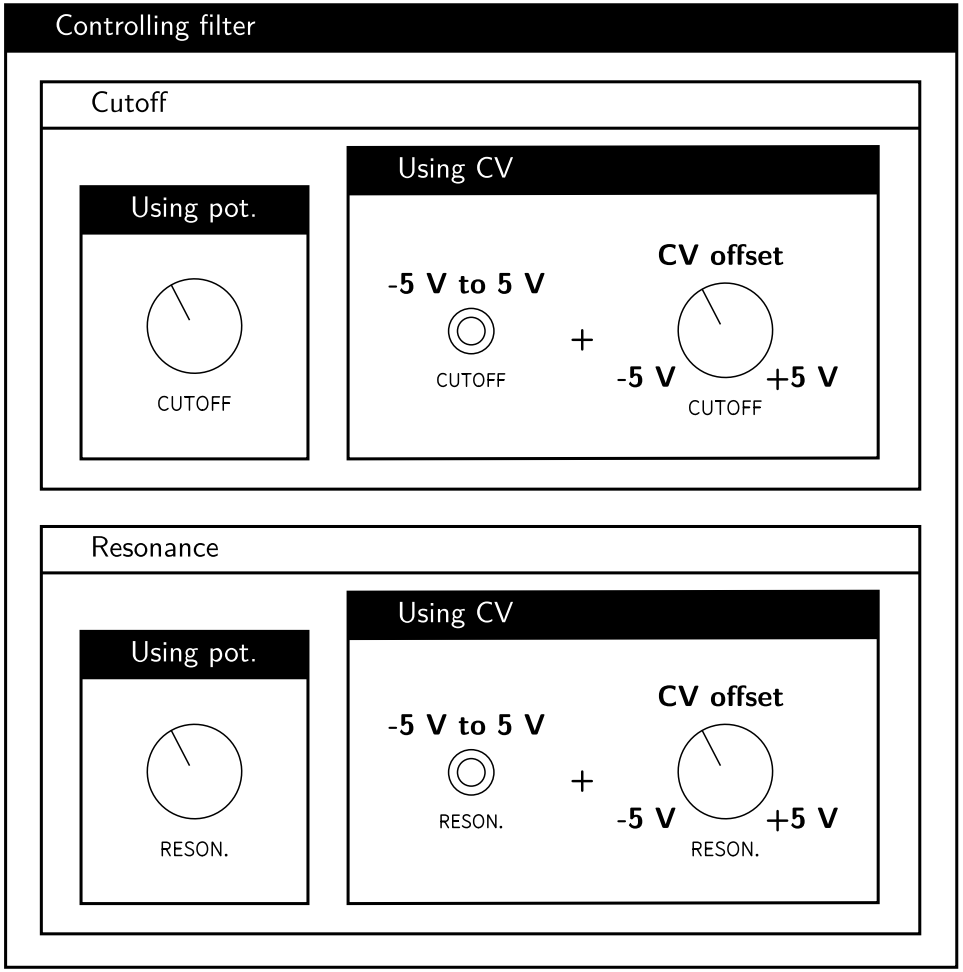


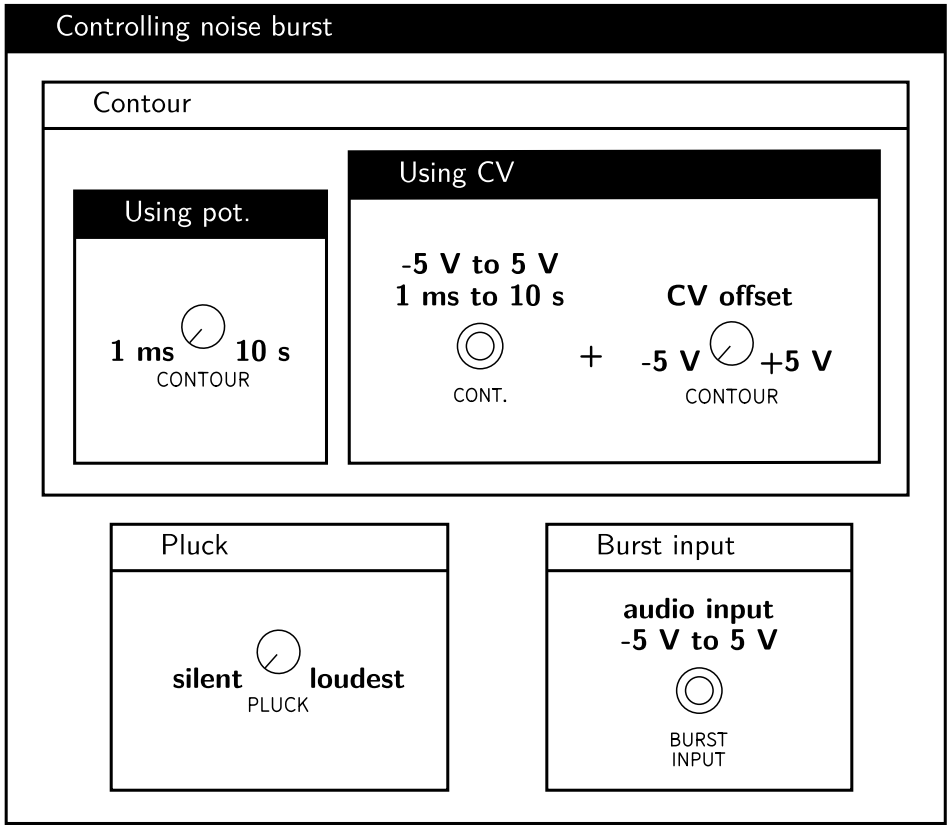


8 String synthesis

The arpeggiator strums up to 8 virtual strings driven by the Karplus-Strong synthesis. Noise burst is internally generated if no BURST input is connected.

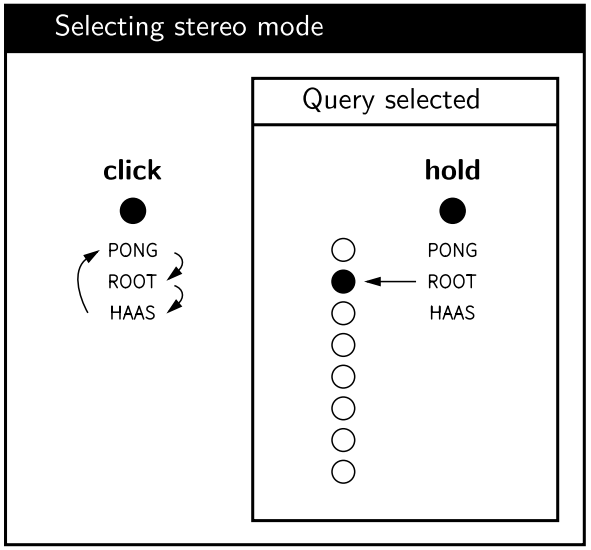


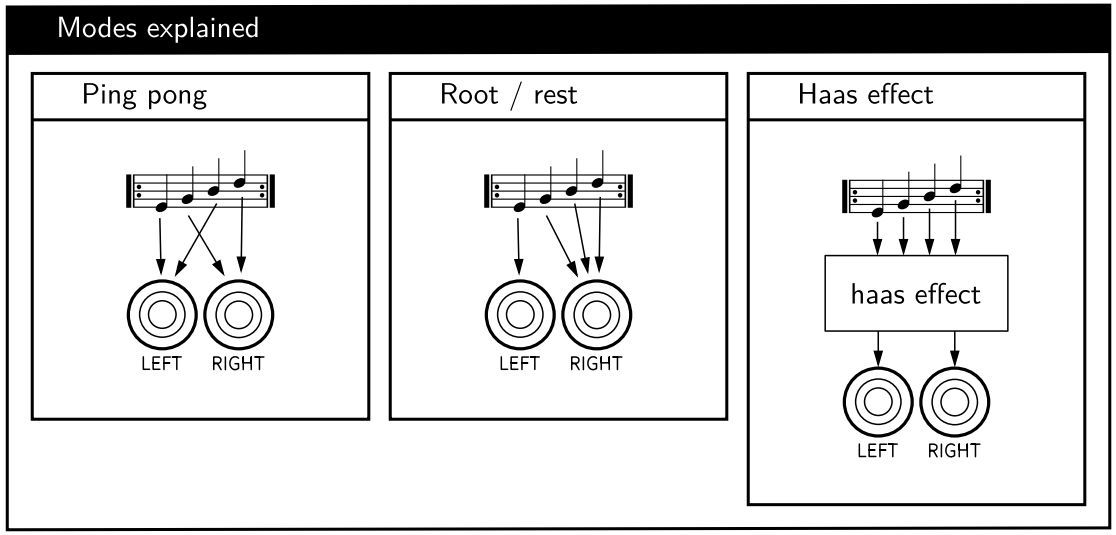


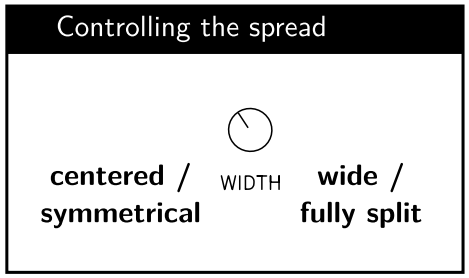


9 Stereo

There are three different modes of routing of the individual strings sound to the audio output.

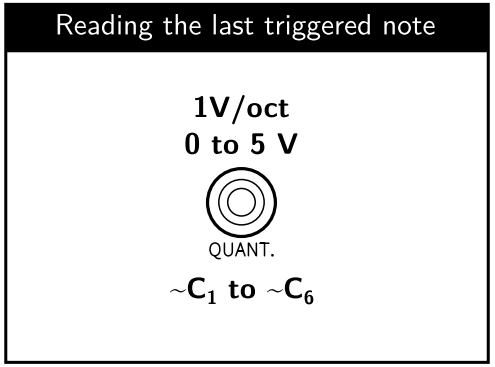






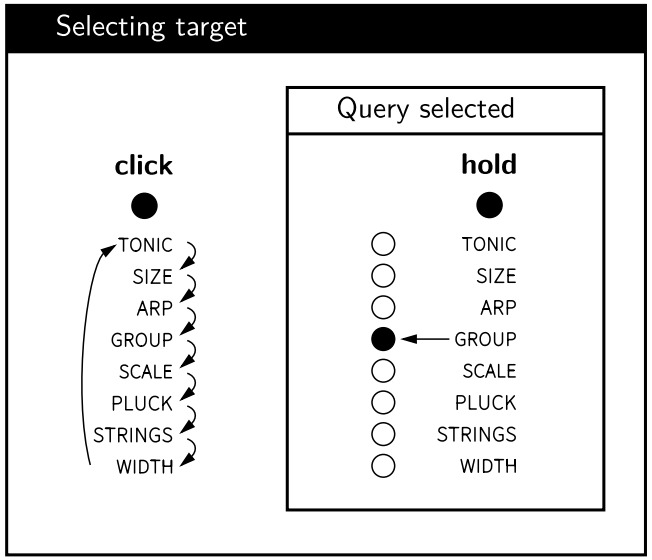
10 Quantized CV output

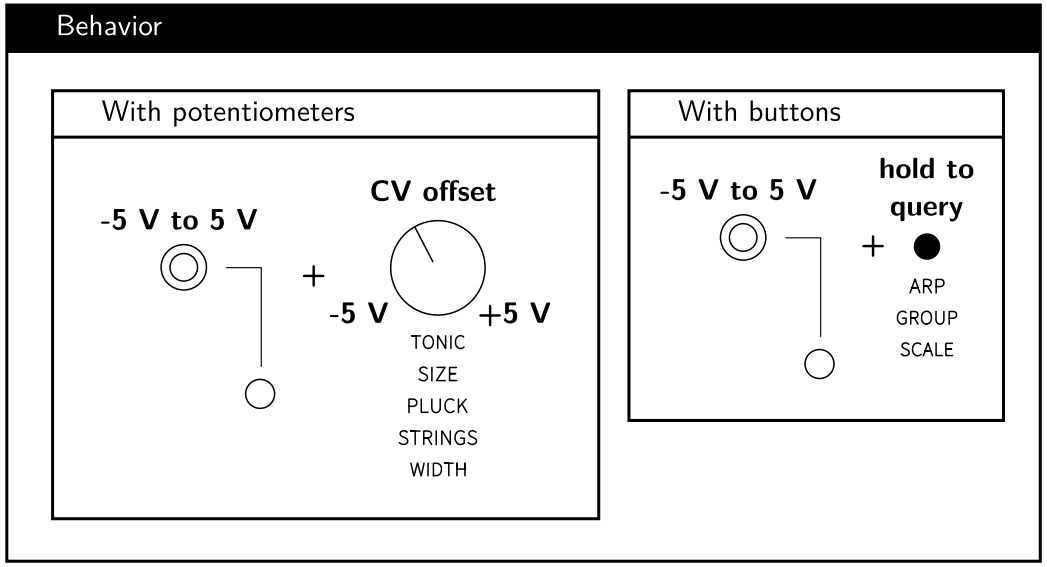
The quantized arpeggiato output used to drive the internal string synthesizer can be also routed out as 1V/oct CV.



11 CV mapping

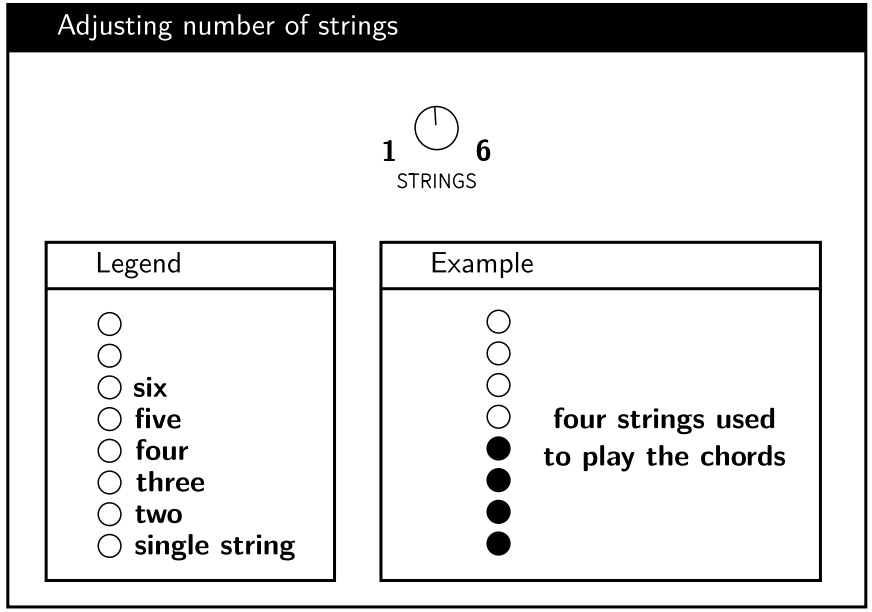
Assign a jack to attributes that otherwise don't have a dedicated CV input.

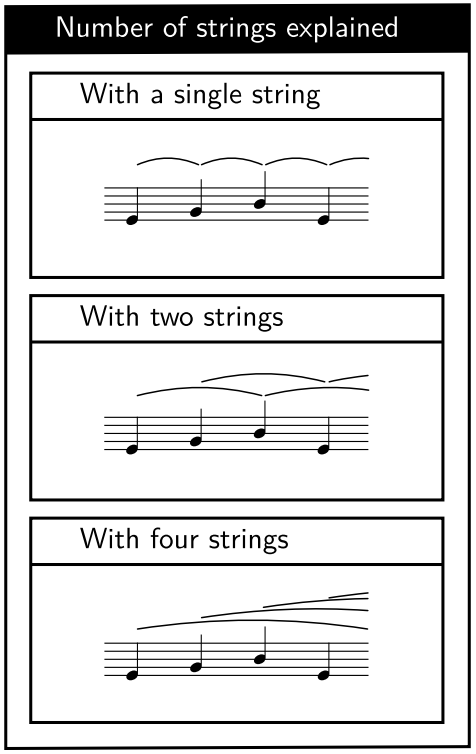




12 Strings

The number of strings used to play the chords.





13 Scale customization

Customize a scale by offsetting its steps by up to 4 quarter tones. During the procedure, the offset is always applied the most recently triggered note.

Configuring offsets

1. Unlock the scale edit mode

hold + **click**

RESET ● GROUP ○ PONG ○ TONIC ●

SCALE ○

2. Select a scale

3. Trigger a note

4. Adjust its offset:

hold

RESET ● GROUP ● **click: quarter up**

SCALE ● **click: quarter down**

5. Repeat 3 and 4 if needed

6. Lock the scale edit mode

hold + **click**

RESET ● GROUP ○ PONG ○ TONIC ●

SCALE ○

Querying the current offset

1. Trigger a note

2. Query the offset:

hold

RESET ●

Legend

without offset {
- ○ **+1 tone**
- ○ **+3 quarter tones**
- ○ **+1 semitone**
- ○ **+1 quarter tone**
- ○ **−1 quarter tone**
- ○ **−1 semitone**
- ○ **−3 quarter tones**
- ○ **−1 tone**

Resetting scale offsets

1. Unlock the scale edit mode

2. Select a scale

3. Reset the steps:

hold + **hold**

RESET ● GROUP ○ PONG ● TONIC ○

SCALE ○

4. Lock the scale edit mode

13 Calibration

To improve tracking of 1V/oct TONE input, calibrate the module:

1. While holding the RESET button, connect a CV source to TONE.
2. Play note C on the CV source and press the RESET button.
3. Play C one octave higher and press the RESET button again.
4. Successful calibration is confirmed by all LEDs litting up.

To improve the accuracy of the 1V/oct QUANT. output, calibrate it against the source that was previously used to calibrate the TONE input:

1. Complete TONE input calibration.
2. Press down the ⟋ (ARP type) button.
3. Connect a cable between the QUANT. output and TONE input.
4. Release the ⟋ (ARP) button.

14 Reset

Calibration, CV mapping and all other attributes are persisted between restarts of the module. To reset their values, hold the RESET button while powering the module on.

15 Questions

Feel free to reach out.

You can also find more information about the module on the website **https://zlosynth.com/arplus**.

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16 Kudos

Kudos to all the eurorack, DSP, and embedded programming communities online. Here are some of the people who helped shape this module:

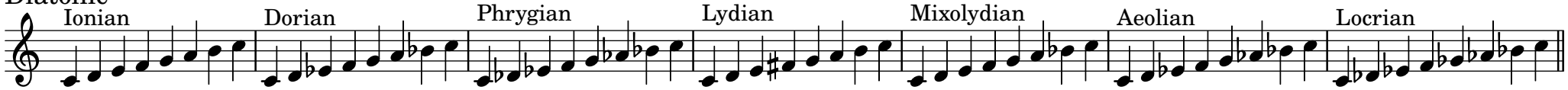
Tomas Niesner, for his extensive testing and thoughtful feedback on the interface, features, and functionality. His insights shaped much of this module.

Jan from Bremen, whose feedback and clever suggestions to the manual of Kaseta (sorry I did not get to implement those) inspired the approach I took with this manual.

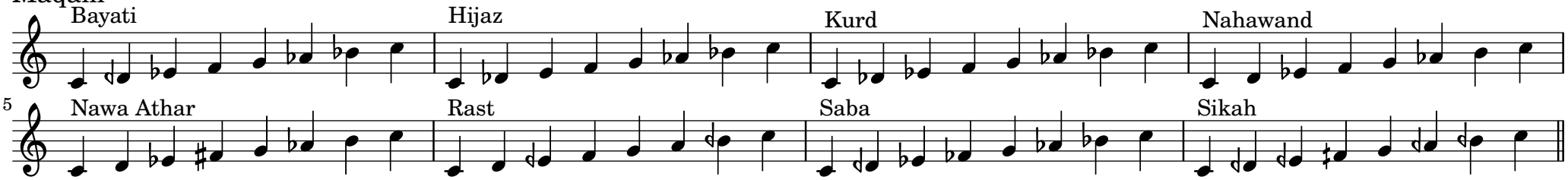
Appendix: Scales

All the available scales divided into groups.

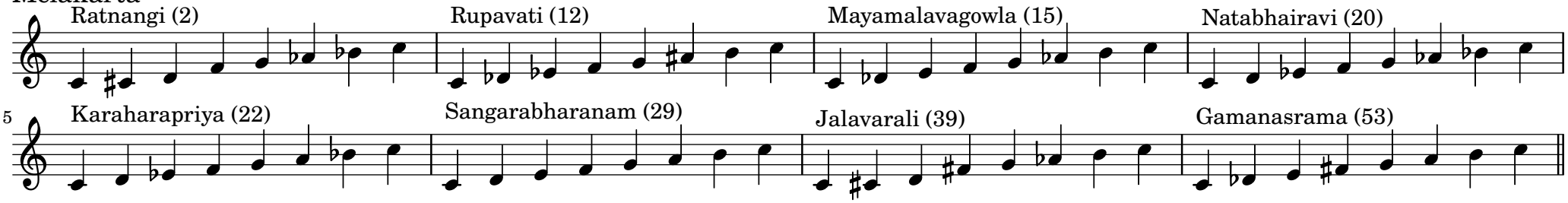
Diatonic



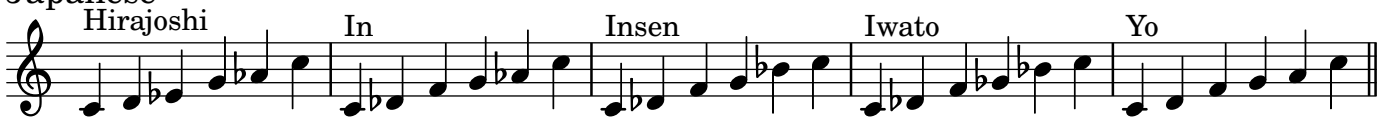
Maqam



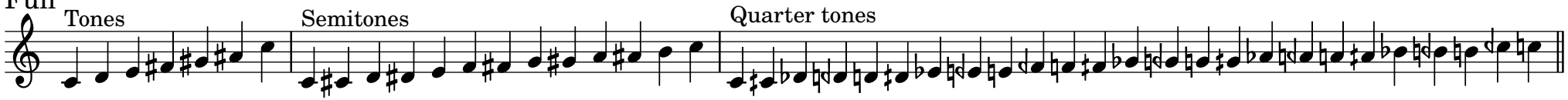
Melakarta



Japanese



Full



Appendix: Chords

All the available chords grouped by size. The actual notes will differ based on the selected scale and tonic.

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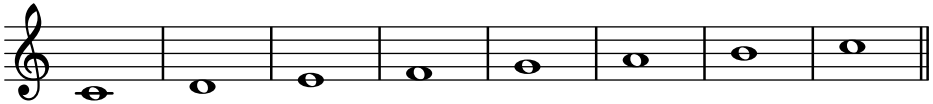

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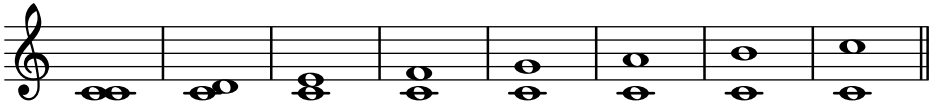
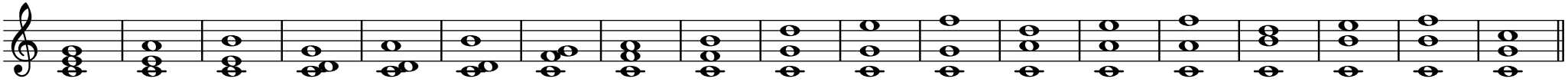


Size: 3



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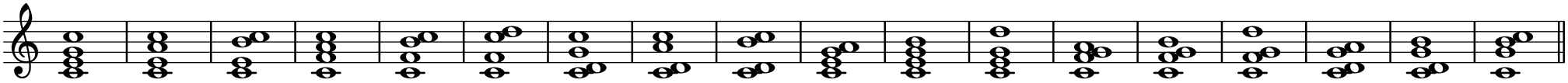

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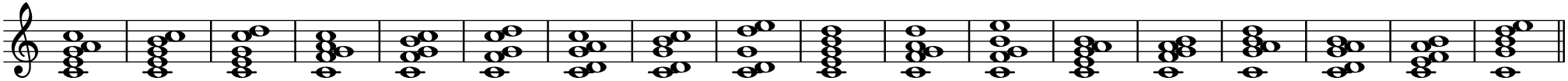

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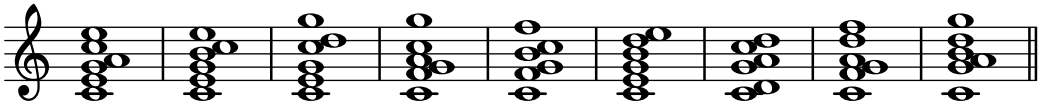

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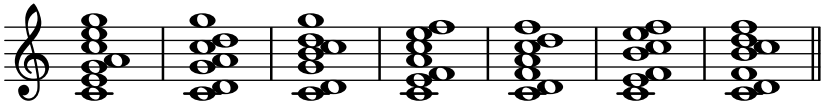
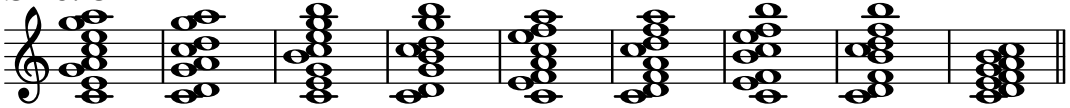
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